

System Design and Architecture

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Document control

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1. Context diagram

mermaid

flowchart LR

Customer[Customer Browser/App]

Admin[Admin Console]

Staff[Restaurant Staff]

Gateway[Payment Gateway]

Email[Email Service]

subgraph TitansInstaFood[Titans InstaFood Platform]

API[Application API]

DB[(Database)]

end

Customer --> API

Admin --> API

Staff --> API

API --> DB

API --> Gateway

API --> Email

Page 3/46 | Controlled copy

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2. Logical ERD

mermaid
erDiagram
Customer ||--o{ Order : places
Customer ||--o{ Payment : makes
Employee ||--o{ Order : processes
Order ||--|{ OrderHasItems : contains
Item ||--o{ OrderHasItems : referenced_by
Order ||--o| Payment : settled_by
Order ||--o| Delivery : fulfilled_by
Menu }o--o{ Item : includes
Employee ||--o{ Delivery : drives
Customer {
int CustomerID PK
string FirstName
string LastName
string Email
string Phone
}
Employee {
int EmployeeID PK
string FirstName
string LastName
string Position
}
Item {
bigint ItemID PK
string Name
decimal Price
boolean Available
string CategoryCode
}
Menu {
int MenuID PK
string MenuName
}
Order {
int OrderID PK
int CustomerID FK
int EmployeeID FK
decimal OrderAmount
string Status
string FulfillmentType
}
OrderHasItems {
int OrderHasItemsID PK
int OrderID FK
bigint ItemID FK

Page 4/46 | Controlled copy

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int Quantity

decimal UnitPrice

}

Payment {

int PaymentID PK

int OrderID FK

decimal PaymentAmount

string PaymentStatus

}

Delivery {

int DeliveryID PK

int OrderID FK

string DeliveryStatus

}

Page 5/46 | Controlled copy

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3. Role model

Role Capabilities

Customer FR1-FR4, FR8-FR20, FR23-FR24, FR17-FR18

Administrator All customer-facing read + FR5-FR7, FR21-FR22, FR25-FR30

Staff Accept/process orders (UC 2.5-2.7), limited PII (DR 2.8)

Driver Delivery status updates (DR 2.6); R2 tracking

System Audit, backup, retention jobs

Page 6/46 | Controlled copy

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4. Recommended logical tiers (R1)

Tier Responsibility

Presentation Responsive web (customer + admin)

API REST or GraphQL; auth JWT/session

Domain Orders, cart, menu, payments orchestration

Integration Payment gateway adapter, email adapter

Data Relational DB; migrations from DDD-Logical-Data-Dictionary-v4.0.pdf

Page 7/46 | Controlled copy

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5. Key integration points

Integration Requirements Notes

Payment gateway FR13, NFR3, NFR10 Webhooks for async confirmation

Email R2 FR32 Order confirmation

Optional maps R2 FR36 Delivery ETA/tracking

Page 8/46 | Controlled copy

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6. API contract (authoritative)

Full REST contract: SDD-0402-Design-API-REST-Contract-Specification-v4.0.pdf

FR-to-endpoint map: API-REST-Contract-Specification-v4.0.pdf

Standards (errors, pagination, auth): ADD-API-Design-Standards-v4.0.pdf

Stack selection is unconstrained; see SDD-System-Design-and-Architecture-v4.0.pdf.

RBAC Permission Matrix

Roles: customer, staff, admin, driver (driver active for delivery features; R2 FR36 extends)

Legend: ? allowed · ? denied · ? masked (read limited fields per DR 2.8)

Customer data

Action	customer	staff	admin	driver
View own profile	?	?	?	?
Update own profile	?	?	?	?
Delete own account	?	?	?	?
View other customer PII	?	?	name+order only	?
			delivery address only	

Menu and catalog

Action	customer	staff	admin	driver
Browse menu (FR24)	?	?	?	?
Search/filter (FR19-20)	?	?	?	?
CRUD menu items (FR5-7)	?	?	?	?
Manage promotions (FR26-27)	?	?	?	?

Cart and orders

Action	customer	staff	admin	driver
Manage own cart (FR8-12)	?	?	?	?
Place order / checkout (FR23)	?	?	?	?
View own order status (FR17)	?	?	?	?
			assigned only	
View own order history (FR18)	?	?	?	?
Accept/process orders (UC 2.5-2.7)	?	?	?	?
Update order status	?	?	?	?
			delivery only	
Cancel order (policy)	?	own, if Pending	?	?

Payments

Action	customer	staff	admin	driver
Pay for own order (FR13)	?	?	?	?
Update payment methods (FR14)	?	?	?	?
View full PAN	?	?	?	?
View payment reports (FR25)	?	?	?	?
Confirm payment (UC 2.6)	?	?	?	?

Reports and reviews

Action	customer	staff	admin	driver
Sales reports (FR21-22)	?	?	?	?
Feedback reports (FR28)	?	?	?	?
Promotion effectiveness (FR29)	?	?	?	?

Page 9/46 | Controlled copy

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| Respond to reviews (FR30) | ? | ? | ? | ? |

| Submit review (R2 FR37) | ? | ? | ? | ? |

System

| Action | customer | staff | admin | driver |

|-----|-----|-----|-----|-----|

RBAC Permission Matrix

Roles: customer, staff, admin, driver (driver for delivery; R2 FR36 extends tracking)

Legend: Y = allowed | N = denied | M = masked (limited fields per DR 2.8)

Customer data

| Action | customer | staff | admin | driver |

| View own profile | Y | N | Y | N |

| Update own profile | Y | N | Y | N |

| Delete own account | Y | N | Y | N |

| View other customer PII | N | M | Y | M |

Menu and catalog

| Browse menu (FR24) | Y | Y | Y | N |

| Search/filter (FR19-20) | Y | Y | Y | N |

| CRUD menu items (FR5-7) | N | N | Y | N |

| Manage promotions (FR26-27) | N | N | Y | N |

Cart and orders

| Manage own cart (FR8-12) | Y | N | N | N |

| Place order / checkout (FR23) | Y | N | Y | N |

| View own order status (FR17) | Y | M | Y | M |

| View own order history (FR18) | Y | N | Y | N |

| Accept/process orders | N | Y | Y | N |

| Update order status | N | Y | Y | Y |

| Cancel order (policy) | Y | N | Y | N |

Payments

| Pay for own order (FR13) | Y | N | N | N |

| Update payment methods (FR14) | Y | N | Y | N |

| View full PAN | N | N | N | N |

| View payment reports (FR25) | N | N | Y | N |

| Confirm payment (UC 2.6) | N | Y | Y | N |

Reports and reviews

| Sales reports (FR21-22) | N | Y | Y | N |

| Promotion effectiveness (FR29) | N | Y | Y | N |

| Feedback reports (FR28) | N | Y | Y | N |

| Respond to reviews (FR30) | N | Y | Y | N |

| Submit review (R2 FR37) | Y | N | Y | N |

API enforcement: Each operation lists x-roles. Middleware returns 403 FORBIDDEN when role not in allowed set.

See API-REST-Contract-Specification-v4.0.pdf for FR-to-endpoint mapping.

1. Order status

Status Description Customer view (FR17) Staff actions
Pending Created; awaiting payment "Awaiting payment" -
Paid Payment captured "Confirmed" Accept ? Preparing
Preparing In kitchen "Preparing" ? Ready
Ready Ready for pickup/dispatch "Ready" Complete or dispatch
OutForDelivery Driver en route "On the way" Driver updates
Completed Fulfilled "Completed" -
Cancelled Not fulfilled "Cancelled" -
Returned Delivery failed "Returned" Admin follow-up
Page 11/46 | Controlled copy
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2. Payment status

Status Maps to order

Initiated Order stays Pending

Successful Order ? Paid

Failed Order stays Pending or ? Cancelled after timeout

Page 12/46 | Controlled copy

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3. Delivery status (entity Delivery)

Status When

Scheduled Order Ready + delivery

OutForDelivery Driver picked up

Delivered Confirmation (DR 2.6)

Returned Failed delivery

Page 13/46 | Controlled copy

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4. Timers (spec defaults)

Timer Duration Action

Pending payment 30 minutes Auto-cancel Pending order

Real-time status SLA 30 seconds FR17 customer refresh

Admin dashboard 10 seconds NFR13

Page 14/46 | Controlled copy

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5. Cart (pre-order)

Cart is not an order status. Cart lines persist until checkout creates Pending order or cart cleared (FR9-FR12).

Integration Specification

External systems for MVP. No SDK code in this repo - contract only.

Page 15/46 | Controlled copy

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1. Payment gateway

Item Specification

Purpose FR13, NFR3, NFR10

Pattern Hosted fields or redirect; no PAN on Titans servers

Stored data GatewayTransactionId, token, last4, brand, expiry month/year

PCI scope SAQ A or A-EP per integration model - see SDD-System-Design-and-Architecture-v4.0.pdf

Page 16/46 | Controlled copy

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1.1 Outbound (platform ? gateway)

Event When

createPaymentIntent Customer checkout

capture Order confirmed

refund Admin-initiated cancel after capture

Page 17/46 | Controlled copy

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1.2 Inbound webhooks (gateway ? platform)

Event type Action

payment.succeeded Mark payment Successful; order ? Paid

payment.failed Mark Failed; notify customer

payment.refunded Audit; adjust reporting

Page 18/46 | Controlled copy

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2. Email service (MVP minimal)

Template Trigger Requirement

registration_verify FR1 signup Verification link, 24h expiry

password_reset R2 FR41 Reset link

order_confirmation R2 FR32 Optional MVP *Should*: send on Paid

Page 19/46 | Controlled copy

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3. Optional R2 integrations

System FR Notes

SMS/push FR45 Provider TBD at R2

Maps/ETA FR35-36 Delivery ETA

CAPTCHA FR48 Login abuse

Page 20/46 | Controlled copy

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4. Failure handling

Dependency down Behavior

Payment gateway 503; queue retry; show user message

Email Log failure; do not block order Paid transition

Webhook processing Dead-letter queue; manual replay tool in ops runbook

Page 21/46 | Controlled copy

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5. Test doubles (implementation repo)

Use gateway sandbox and mail catcher in non-prod per QAP-Quality-Assurance-Plan-v4.0.pdf.

PCI and Security Boundary

Supports FR13, NFR1-NFR3, DR 2.8, NFSR1-5, SAR1-3.

Page 22/46 | Controlled copy

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1. Trust boundaries

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flowchart TB

subgraph public [Public Internet]

Browser[Customer Browser]

end

subgraph dmz [Application Tier]

API[Titans API]

end

subgraph data [Data Tier]

DB[(Database)]

end

subgraph third [Third Party]

GW[Payment Gateway]

end

Browser -->|HTTPS JSON| API

API -->|TLS token only| GW

API -->|encrypted| DB

Browser -. ->|card fields iframe/hosted| GW

Cardholder data environment (CDE): Minimize - prefer gateway-hosted collection so Titans API never receives PAN.

Page 23/46 | Controlled copy

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2. Data classification

Class Examples Storage Staff access

Public Menu, prices DB All

Internal Order IDs, status DB staff, admin

Confidential Email, phone, address DB encrypted at rest staff masked, admin full

Restricted Payment tokens, auth secrets DB/vault admin ops only

Prohibited on Titans Full PAN, CVV, magnetic data Never -

Page 24/46 | Controlled copy

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3. DR 2.8 - front attendant masking

Staff role must not API-return: passwordHash, full PAN, SSN, full email (show j*@domain.com optional).

Page 25/46 | Controlled copy

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4. Controls checklist (MVP spec)

Control Requirement

TLS 1.2+ NFSR1

MFA for admin NFR1

RBAC SDD-System-Design-and-Architecture-v4.0.pdf

Audit log NFR2, SAR2

Input validation NFSR5, API schemas

Rate limiting login AUTH-01, API_STANDARDS

Retention jobs DRRD1-7

Page 26/46 | Controlled copy

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5. Legal

SAR3 references GDPR/HIPAA - legal counsel must confirm applicability before marketing compliance. This document is technical scoping only.

Page 27/46 | Controlled copy

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6. Security testing (evidence in implementation phase)

Test Plan reference

OWASP ZAP baseline QAP-Quality-Assurance-Plan-v4.0.pdf

Dependency scan Implementation CI

Pen test Pre-production gate

Page 28/46 | Controlled copy

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1. Production topology

mermaid

flowchart TB

users[Users]

cdn[CDN / WAF]

web[Web App Static or SSR]

api[API Service x2+]

db[(Primary DB)]

replica[(Read Replica optional)]

backup[(Backup Storage)]

gw[Payment Gateway]

users --> cdn --> web

web --> api

api --> db

api --> replica

db --> backup

api --> gw

Page 29/46 | Controlled copy

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2. Component requirements

Component Minimum

API instances 2+ for availability (NFR5)

Database Managed relational with automated backup (DBRRD1, NFR6)

TLS Terminated at load balancer or CDN

Secrets Vault / cloud secret manager

Logs Centralized; retention per SAR2

Page 30/46 | Controlled copy

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3. Regions

Tier Recommendation

MVP Single region acceptable

DR Off-site backup; warm standby for NFR21

Page 31/46 | Controlled copy

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4. Release process (implementation repo)

Page 32/46 | Controlled copy

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1. Deploy staging ? full regression (Application-Test-Plan.csv)

Page 33/46 | Controlled copy

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2. NFR spot checks per QAP-Quality-Assurance-Plan-v4.0.pdf

Page 34/46 | Controlled copy

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3. Sponsor approval (Implementation-Readiness-Checklist-v4.0.pdf §E)

Page 35/46 | Controlled copy

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4. Production deploy with maintenance window if needed

Page 36/46 | Controlled copy

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5. Rollback

Trigger Action

Error rate > 5% post-deploy Roll back API + web to previous version

DB migration failure Restore from pre-migration backup

Page 37/46 | Controlled copy

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1. Environments

Environment Purpose Data

local Developer workstations Synthetic seed

dev Integration Synthetic + gateway sandbox

staging UAT / QA Anonymized copy or seed

production Live customers Real data

Page 38/46 | Controlled copy

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2. Required configuration variables

Variable Required Description

APP_ENV Yes local \

API_BASE_URL Yes Public API URL

DATABASE_URL Yes Connection string (secrets manager in prod)

JWT_SECRET Yes Signing key (rotate per policy)

JWT_ACCESS_TTL_MIN No Default 15

JWT_REFRESH_TTL_DAYS No Default 7

PAYMENT_GATEWAY_API_KEY Yes Sandbox vs prod key

PAYMENT_WEBHOOK_SECRET Yes Webhook HMAC

PAYMENT_GATEWAY_MODE Yes sandbox \

EMAIL_API_URL Staging+ Email provider

EMAIL_API_KEY Staging+ Secret

EMAIL_FROM Yes orders@titansinstafood.example

CORS_ALLOWED_ORIGINS Yes Comma-separated web origins

RATE_LIMIT_LOGIN_MAX No Default 5 per 15 min

ORDER_PAYMENT_TIMEOUT_MIN No Default 30

Page 39/46 | Controlled copy

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3. Feature flags (MVP)

Flag Default FR

FEATURE_EMAIL_VERIFICATION true FR1

FEATURE_SAVED_CART true FR11

FEATURE_DELIVERY false Enable when delivery scope approved

FEATURE_R2_GUEST_CHECKOUT false FR31

Page 40/46 | Controlled copy

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4. Non-production rules

- PAYMENT_GATEWAY_MODE must not be live in local/dev
- Production secrets only via vault / CI secret store - never committed

Page 41/46 | Controlled copy

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5. Health endpoints (implementation)

Path Use

GET /health/live Liveness

GET /health/ready DB + gateway reachability

Page 42/46 | Controlled copy

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1. Mandatory constraints

ID Constraint Rationale

TC-1 Relational database for transactional data FR16, ACID orders/payments

TC-2 REST API conforming to SDLC/04-Design/API-REST-Contract-Specification-v4.0

Interop, contract

TC-3 HTTPS everywhere NFR3

TC-4 Password hashing (Argon2id or bcrypt cost ≥ 12) NFSR3

TC-5 Parameterized queries / ORM - no string-concat SQLNFSR5

TC-6 Secrets outside source control Security

TC-7 UTC timestamps in persistence API_STANDARDS

TC-8 Decimal type for money in DB DATA_DICTIONARY

Page 43/46 | Controlled copy

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2. Recommended (non-binding)

Area Recommendation

API implementation Any language with OpenAPI 3 support

Frontend SPA or SSR with responsive CSS

Hosting Cloud with managed DB and TLS termination

Migrations Versioned schema migration tool in implementation repo

Page 44/46 | Controlled copy

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3. Prohibited

Item Reason

Storing full PAN/CVV PCI

Logging passwords or tokens Security

FR15 (retired requirement) Duplicate removed

Page 45/46 | Controlled copy

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4. Implementation repository expectations

Deliverable Not in this spec repo

Source code Yes - separate repo

CI/CD pipelines Yes

Infrastructure-as-code Yes

Automated tests executing against running app Yes

Page 46/46 | Controlled copy